

भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 360: Machine Learning Lab
Practice Assignment 8

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Consider the **Heart Disease dataset** to classify patients having heart disease or not based on the given predictors.
   Write a program to do the followings:
   (a) Read the dataset.
   (b) Perform feature scaling / normalization, if necessary.
   (c) Implement K-NN classifier algorithm using $K = 3, 5, 7$ to classify the patients.
   (d) Evaluate the designed classifier using 5-fold cross-validation. Report the results in terms accuracy, class-wise accuracy, precision and recall for all three $k-$values, separately.
   (e) Re-implement 1(c) and 1(d) using weighted K-NN classifier using the inverse-distance weighting formula:
   $w_i = 1/(d_i + \epsilon)$; where d_i is the distance of the i-th neighbor and ϵ is a small positive constant, like 0.01.
   (f) Compare the predictions with those of standard K-NN, and report your observations.

2. Consider the **Diabetes dataset** for classifying patients to be Diabetic, Non-Diabetic, or Predict-Diabetic, based on the given predictors.
   (a) Read the dataset.
   (b) Implement the ID3 decision tree algorithm using the followings:
      1. Information gain to select the best splitting attribute.
      2. Thresholding to convert continuous attributes-values to categorical values.

      Stopping conditions:

      - Pure subsets are obtained (i.e., every observation in the current subset belongs to the same target class).
      - Attribute Exhaustion (i.e., no more unused attributes left to split the data).

   (c) Evaluate the designed classifier using 5-fold cross-validation. Report the results in terms accuracy, class-wise accuracy, precision and recall.